

DEFINITE INTEGRALS

Introduction

The definite integrals can be evaluated by finding any antiderivative of the integrand and then subtracting the value of this antiderivative at the lower limit of integration from its value at the upper limit of integration.

Theorem (The Fundamental Theorem of Calculus)

If f is continuous on $[a, b]$ and if F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a)$$

Example 1

Evaluate $\int_0^3 f(x) dx$ given that $f(x) = \begin{cases} x^2, & x < 2 \\ 3x - 2, & x \geq 2 \end{cases}$

Solution: Write

$$\begin{aligned} \int_0^3 f(x) dx &= \int_0^2 f(x) dx + \int_2^3 f(x) dx = \int_0^2 x^2 dx + \int_2^3 (3x - 2) dx \\ &= \left[\frac{x^3}{3} \right]_0^2 + \left[\frac{3x^2}{2} - 2x \right]_2^3 = \left(\frac{8}{3} - 0 \right) + \left(\frac{27}{2} - 4 \right) - (6 - 4) = \frac{49}{6} \end{aligned}$$

Area Formula (Type I Region)

The area A of the region S bounded by the curves $y = f(x)$, $y = g(x)$, and the lines $x = a$, $x = b$, where f and g are continuous functions and $f(x) \geq g(x)$ for all x in $[a, b]$, is given by the formula

$$A = \int_a^b [f(x) - g(x)] dx$$

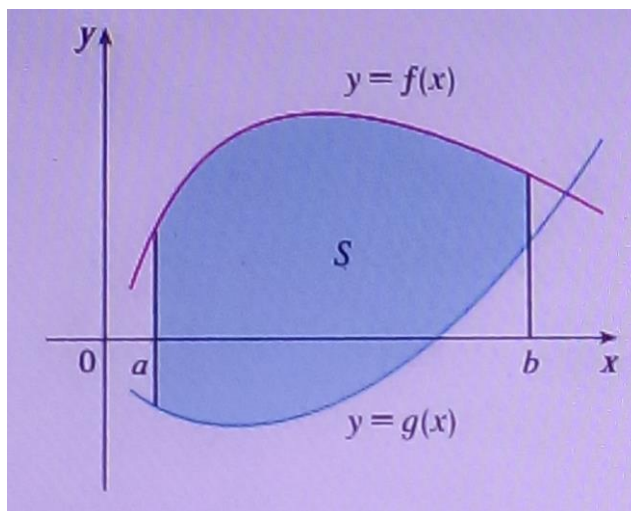


Figure 1

Example 2

Find the area of the region enclosed by the parabolas $y = x^2$ and $y = 2x - x^2$.

Solution: We first find the points of intersection of the parabolas by solving their equations simultaneously. This gives $x^2 = 2x - x^2$ or $2x^2 - 2x = 0$. Thus we get $2x(x - 1) = 0$ which gives $x = 0$ or $x = 1$. Therefore, the points of intersection are $(0, 0)$ and $(1, 1)$.

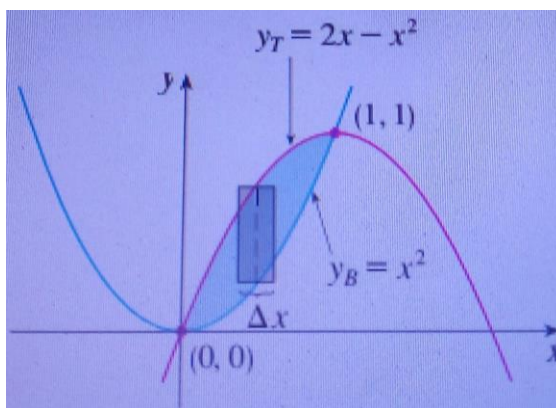


Figure 2

Using area formula, the desired area is

$$\begin{aligned}\int_0^1 [(2x-x^2)-x^2] dx &= \int_0^1 (2x-2x^2) dx = [x^2]_0^1 - \frac{2}{3}[x^3]_0^1 \\ &= [1-0] - \frac{2}{3}[1-0] = 1 - \frac{2}{3} = \frac{1}{3}.\end{aligned}$$

Area Formula (Type II)

The area A of the region bounded by the curves $x=f(y)$, $x=g(y)$, and the lines $y=c$, $y=d$, where f and g are continuous functions and $f(y) \geq g(y)$ for $c \leq y \leq d$, is given by the formula

$$A = \int_c^d [f(y) - g(y)] dy$$

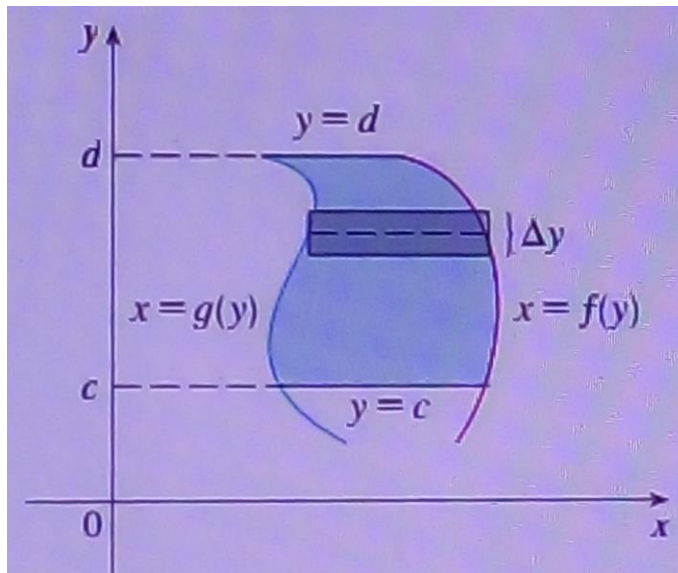


Figure 3

Example 3

Find the area enclosed by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$.

Solution: By solving the two equations we find that the points of intersection are $(-1, -2)$ and $(5, 4)$.

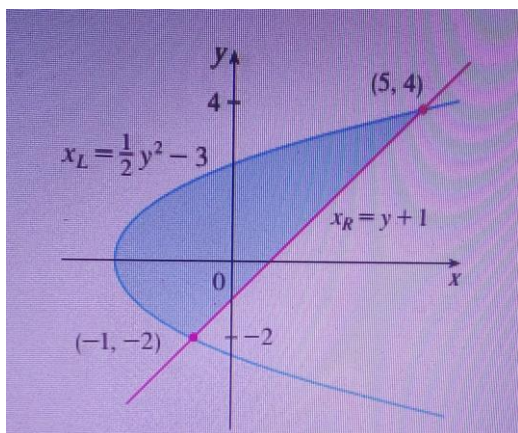


Figure 4

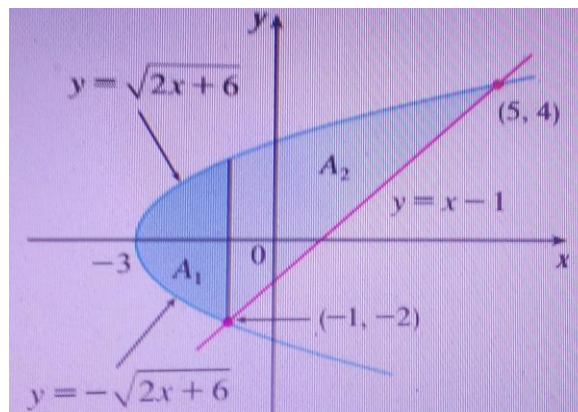


Figure 5

Using area formula, the desired area is given by (using Figure 4)

$$\begin{aligned}
 A &= \int_{-2}^4 \left[(y+1) - \left(\frac{1}{2}y^2 - 3 \right) \right] dy = \left[\frac{1}{2}y^2 + y - \frac{1}{6}y^3 + 3y \right]_{-2}^4 \\
 &= \left[\frac{1}{2} \times 16 + 4 - \frac{1}{6} \times 4^3 + 3 \times 4 \right] - \left[\frac{1}{2} \times 4 - 2 + \frac{8}{6} - 6 \right] \\
 &= \left[\frac{144 - 64}{6} \right] - \left[\frac{-28}{6} \right] = \frac{80 + 28}{6} = 18
 \end{aligned}$$

Exercises

Sketch the region enclosed by the following curves and find the area:

1. $y = x^2 + 1$, $y = x$, $x = -1$, $x = 2$

2. $y = \sqrt{x}$, $y = -\frac{1}{4}x$, $x = 0$, $x = 4$

3. $y = 2x$, $y = x^2$

4. $y^2 = 4x$, $y = 2x - 4$

5. $y = 5x - x^2$, $y = x$

6. $x = y^2 - 4y$, $x = 2y - y^2$